# Lecture 02, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

# General Solutions of common ODEs

$y' = ay$; a general solution is $y(x) = A\mathrm{e}^{ax}$

$y' + a^2 y = 0$; a general solution is $y(x) = A\cos(ax) + B\sin(ax)$

$y' - a^2 y = 0$; a general solution is $y(x) = A\mathrm{e}^{ax} + B\mathrm{e}^{-ax}$

where $a \in \mathbb{R}$ is a constant and $A$ and $B$ are free parameters.

Ex: given the equation $y'' + 3y = 0$ a general solution is

$$y(x) = A\cos(\sqrt{3}x) + B\sin(\sqrt{3}x)$$

To verify it is a solution, take the double derivative

$$y'' = -3A\cos(\sqrt{3}x) - 3B\sin(\sqrt{3}x)$$

and put back into the equation:

$$y'' + 3y =$$

$$-3A\cos(\sqrt{3}x) - 3B\sin(\sqrt{3}x) + 3A\cos(\sqrt{3}x) + 3B\sin(\sqrt{3}x) = 0$$

# Initial Value Problem

Ex. of an IVP (initial value problem), an ODE with an initial condition (IC)

$$\begin{cases} y' = -5y & \leftarrow \text{ODE} \\ y(0) = 100 & \leftarrow \text{IC} \end{cases}$$

To solve we find a general solution to the ODE, $y(x) = A\,\mathrm{e}^{-5x}$, and then apply the IC: $y(0) = A = 100$ so $A = 100$ and the solution to the IVP is $\boxed{y(x) = 100\,\mathrm{e}^{-5x}}$

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# 1st Order Linear ODE

Consider the 1st order linear equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\sin(x)}{\cos(x)}y = \frac{x}{\cos(x)}$$

Multiply by $\cos(x)$

$$\cos(x)\frac{\mathrm{d}y}{\mathrm{d}x} - \sin(x)y = x \;\rightarrow\; \frac{\mathrm{d}}{\mathrm{d}x}(\cos(x)y) = x$$

Integrate both side in $x$:

$$\cos(x)y = \frac{x^2}{2} + A$$

Therefore

$$y(x) = \frac{x^2}{2\cos(x)} + \frac{A}{\cos(x)}$$

Can this be done for any 1st order linear equation?

# Integrating Factor

Consider a generic 1st order linear ODE for $y(x)$:

$$y' + p(x)y = q(x)$$

We multiply by a function $\mu(x)$ and rewrite:

$$\mu y' + p\mu y = \mu q \text{ and we want } (\mu y)' = \mu q$$

which is easy to solve (just integrate both sides)

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We need $\underline{\mu y'} + p\mu y = (\mu y)' = \mu' y + \underline{\mu y'}$

and specifically $p\mu y = \mu' y$ , in other words $\mu' = p\mu$.

To solve this we separate and integrate both sides:

$$\frac{\mathrm{d}\mu}{\mathrm{d}x} = \mu p(x) \rightarrow \frac{\mathrm{d}\mu}{\mu} = p\mathrm{d}x$$

$$\int \frac{\mathrm{d}\mu}{\mu} = \int p(x)\mathrm{d}x \rightarrow \ln(|\mu|) = \int p(x)\mathrm{d}x$$

$$\boxed{\mu(x) = \mathrm{e}^{\int p\mathrm{d}x}}$$

The function $\mu$ is called integrating factor

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The previous example was

$$y' - \frac{\sin(x)}{\cos(x)} y = \frac{x}{\cos(x)}$$

In this case

$$p(x) = -\frac{\sin(x)}{\cos(x)}$$

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Consider the IVP:

$$\begin{cases} xy' = 3y + x^2 \\ y(1) = 0 \end{cases}$$

Rewrite the equation as $y' = \frac{3}{x}y + x$ and then $y' - \frac{3}{x}y = x$ (*)

So I.F. is $\mu(x) = \mathrm{e}^{-\int \frac{3}{x}\mathrm{d}x} = \mathrm{e}^{-3\ln(x)} = \mathrm{e}^{\ln(x^{-3})} = x^{-3} = \frac{1}{x^3}$

Multiply (*) by $\mu$ to get

$$\frac{y'}{x^3} - \frac{3}{x^4}y = \frac{1}{x^2} \quad \rightarrow \quad \left(\frac{1}{x^3}y\right)' = \frac{1}{x^2}$$

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Integrate both sides in $x$:

$$\frac{1}{x^3}y = -\frac{1}{x} + A \quad \rightarrow \quad y = -x^2 + Ax^3 \quad \text{(general solution)}$$

Apply the initial condition:

$$y(1) = -1 + A = 0 \quad \rightarrow \quad A = 1$$

So the solution to the IVP is $\boxed{\text{y(x)} = \text{-x}^2 + x^3}$

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Another ex: solve the IVP: $y' + 2xy = 1; \qquad y(1) = 1$

We can calculate the integrating factor right away:

$\mu = \mathrm{e}^{\int 2x\mathrm{d}x} = \mathrm{e}^{x^2}$

Multiply the equation by $\mu$:

$$y'\,\mathrm{e}^{x^2} + 2x\mathrm{e}^{x^2}y = \mathrm{e}^{x^2} \quad \rightarrow \quad \left(\mathrm{e}^{x^2}y\right)' = \mathrm{e}^{x^2}$$

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Integrate: $\mathrm{e}^{x^2} y(x) = \int \mathrm{e}^{x^2} \mathrm{d}x$

Rewrite as $\mathrm{e}^{x^2} y(x) = \int_a^x \mathrm{e}^{s^2} \mathrm{d}s + A$ where we use $s$ as the variable

of integration as the integral cannot be done explicitly. So

$y(x) = \mathrm{e}^{-x^2} \int_a^x \mathrm{e}^{s^2} \mathrm{d}s + A\mathrm{e}^{-x^2}$

It's convenient to pick $a$, which is arbitrary so far, as the point

where the IC is given ($x = 1$):

$y(x) = \mathrm{e}^{-x^2} \int_1^x \mathrm{e}^{s^2} \mathrm{d}s + A\mathrm{e}^{-x^2}$ (general solution)

Note that when we integrate what we are really doing is:

$\int x^2 \mathrm{d}x = \int_a^x s^2 \mathrm{d}s = \frac{s^3}{3}|_a^x = \frac{x^3}{3} + A$ and that $\int_0^x + A = \int_0^1 + \int_1^x + A$

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Apply the initial condition:

$$y(1) = \mathrm{e}^{-1} \cancel{\int_1^1 \mathrm{e}^{s^2} \mathrm{d}s} + A\mathrm{e}^{-1} = A\mathrm{e}^{-1} = 1$$

So $A = \mathrm{e}$ and the solution to the initial value problem is

$$y(x) = \mathrm{e}^{-x^2} \int_1^x \mathrm{e}^{s^2} \mathrm{d}s + A\mathrm{e}^{-x^2} \quad \rightarrow \boxed{\mathrm{y(x)} = \mathrm{e}^{-x^2} \int_1^x \mathrm{e}^{s^2} \mathrm{d}s + \mathrm{e}^{1-x^2}}$$

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# Variation of Parameters

Let's solve the inhomogeneous equation $y' + 2xy = 1$ in a different way.

First we solve the homogeneous equation $y' + 2xy = 0$ by separating: $\frac{\mathrm{d}y}{\mathrm{d}x} = -2xy$

$\rightarrow \int \frac{\mathrm{d}y}{y} = \int -2x\,\mathrm{d}x \quad \rightarrow \quad \ln(y) = -x^2 + A$

So that $y(x) = \mathrm{e}^{-x^2+A} = C\mathrm{e}^{-x^2}$ where $C = \mathrm{e}^A$

So $y = Ce^{-x^2}$ (solution to the homogeneous equation)

Variation of Parameter(s):

Now to solve the inhomogeneous equation by trying a solution of

the form $y(x) = f(x)\,e^{-x^2}$

and notice $y' = f'e^{-x^2} - 2xfe^{-x^2}$

where $f(x)$ is an arbitrary function (we changed $C$ into $f(x)$)

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Put y(x) and y' into the original inhomogeneous equation

$y' + 2xy = 1$:

$$f' e^{-x^2} - \cancel{2xf e^{-x^2}} + \cancel{2xf e^{-x^2}} = 1$$

$$f' e^{-x^2} = 1 \quad \rightarrow \quad f' = e^{x^2}$$

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Integrate to get

$$f(x) = \int_a^x \mathrm{e}^{s^2}\,\mathrm{d}s + A$$

Substituting this back in $y(x) = f(x)\mathrm{e}^{-x^2}$ we get

$$\boxed{\mathrm{y(x)} = \mathrm{e}^{-x^2}\int_a^x \mathrm{e}^{s^2}\,\mathrm{d}s + A\mathrm{e}^{-x^2}}$$

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